

The Arrow of Time: 0/0 of Entropy

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The Removable Singularity Framework

Abstract

We show that the arrow of time is the 0/0 of entropy: time flows because entropy increases, and the 0/0 is at the Big Bang where entropy was extremely low. Boltzmann entropy $S = k_B \ln(\Omega)$ is the fundamental measure. The second law ($dS/dt \geq 0$) is statistical, not absolute. The Past Hypothesis states the Big Bang had low entropy (the cosmic 0/0). Landauer's principle connects information to entropy. All arrows of time (thermodynamic, causal, psychological, cosmic, radiative) emerge from this single 0/0.

1. Boltzmann Entropy

$$S = k_B * \ln(\Omega)$$

Ω is the number of microstates consistent with a macrostate. More microstates = higher entropy. The logarithmic scaling means entropy grows slowly with system size, but for macroscopic systems ($N \sim 10^{23}$), Ω is astronomically large.

2. Second Law of Thermodynamics

$$dS/dt \geq 0$$

The second law is NOT fundamental -- it is statistical. Disordered states have exponentially more microstates than ordered states. Systems naturally evolve toward more probable (higher entropy) states. Time flows because we move toward higher entropy.

3. Past Hypothesis (Cosmic 0/0)

The Big Bang had extremely low entropy ($S \sim 0$). This is the 0/0 of time. The universe started in an incredibly ordered state, and entropy has been increasing ever since (by a factor of 10^{88}). This is the deepest mystery in physics: WHY was entropy low at the Big Bang?

4. Fluctuation Theorem

$$P(+S) / P(-S) = \exp(S / k_B)$$

The fluctuation theorem shows that entropy CAN decrease, but only for small systems and short times. For macroscopic systems, the probability of entropy decrease is effectively zero. Time's arrow is statistical, not absolute.

5. Landauer's Principle

$$E = n * k_B * T * \ln(2)$$

Erasing 1 bit of information creates $kT \ln(2)$ of heat. Information IS physical. This connects information theory to thermodynamics, and supports the holographic principle (Ch.47).

6. Arrows of Time

All arrows of time point the SAME direction:

- Thermodynamic: entropy increases
- Causal: causes precede effects
- Psychological: we remember past, not future
- Cosmic: universe expands
- Radiative: waves go out, not in

ALL emerge from the low-entropy Big Bang (the 0/0).

7. Connections

System	Prior Chapter	Connection
Black holes	Ch.32 Black holes	Hawking radiation
Entanglement	Ch.33 Entanglement	Decoherence
Consciousness	Ch.34 Consciousness	Subjective time
Holographic	Ch.47 Holographic	Info-time
Cosmic web	Ch.46 Cosmic web	Expansion
SOC	Ch.41 SOC	Local order

8. Conclusion

The arrow of time is the 0/0 of entropy: time emerges from the low-entropy Big Bang. The second law is statistical, not absolute. Landauer's principle connects information to entropy. All arrows of time point the same direction. This is the most universal phenomenon in physics, connecting black holes, entanglement, consciousness, holography, and the cosmic web.

References

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